

SIMATS ENGINEERING

Department of Computer Science and Engineering

CSA15 Cloud Computing and Big Data Analytics

CO2 AT1 - Capstone Infrastructure Sizing Workbook

Guided calibration workbook for VM sizing, virtualization decisions, snapshot planning and VM-container comparison



Assessment Metadata

Course Code	CSA15
Course Title	Cloud Computing and Big Data Analytics
Assessment	CO2 AT1 - Virtualization Scenario Problem-Solving
Submission Type	Individual digital workbook based on the same capstone title used in CO1
Faculty	Dr. C. Rajesh Babu
Employee ID	SSETSCS-415
Submission Mode	Completed workbook and evidence repository link through Google Classroom

How This Workbook Works

- Read each guidance block and immediately fill the related response table.
- Use the same capstone title selected for CO1.
- Make reasonable assumptions when exact values are unknown. Good assumptions must be written clearly.
- The goal is not to guess a perfect industry configuration; the goal is to reason technically and justify the first infrastructure plan.
- The final decisions from this workbook will be used again in CO2 AT2 and CO2 AT3.

Technical Tip

A good infrastructure estimate is transparent. If a number is assumed, write why it is assumed. If a number is calculated, show the formula. If a service is selected, justify why it fits the workload.

Student and Capstone Details

Student Name	
Register Number	
Class / Slot	Slot B
Capstone Title	
Capstone Product Type	Web app / Android app / iOS app / Hybrid app

CO1 Architecture Reference

Mention concept map / presentation title / GitHub link if available

Activity 1: Bring Forward CO1 Capstone Context

Start with the cloud product idea already used in CO1. The infrastructure plan must support that same application, not a new unrelated example.

Capstone Element	Student Entry
Primary users	
User roles	
Main modules	
Cloud services identified in CO1	
API/backend need	
Database need	
File upload/storage need	
Analytics/AI/Python module	

Activity 2: Workload Classification

Classify the capstone workload before selecting a VM. A form-based SaaS app and an AI analytics app should not receive the same infrastructure plan.

Workload Type	Typical Signals	Starting VM Guidance
Light web / form app	Login, CRUD, simple reports, low file upload	1-2 vCPU, 2-4 GB RAM
API backend	Authentication, REST APIs, moderate concurrent requests	2 vCPU, 4-8 GB RAM
Database-heavy app	Search, many records, frequent writes	2-4 vCPU, 8-16 GB RAM
File-heavy app	Images, PDFs, uploads, downloads	2 vCPU, 4-8 GB RAM + separate object storage
Analytics app	Dashboards, aggregation, batch reports	4 vCPU, 16 GB RAM
AI/Python module	Prediction, NLP, image/text processing	4-8 vCPU, 16-32 GB RAM; GPU optional
High traffic app	Large user base and peak demand	Multiple VMs or autoscaling group

Technical Tip

Start small for prototype, but do not hide growth. A valid answer can say: prototype uses one VM, pilot separates database, production uses load balancing and managed services.

Student Decision: My capstone workload type and reason

Activity 3: User and Traffic Calibration

Use the following formulas to convert user assumptions into approximate traffic.

Metric	Formula	Example
Concurrent Active Users	Registered Users x Active User % x Peak Factor	500 x 10% x 2 = 100
Requests per Minute	Concurrent Active Users x Actions per User per Minute	100 x 2 = 200
Requests per Second	Requests per Minute / 60	200 / 60 = 3.33 RPS
Daily Requests	Daily Active Users x Avg Actions per Day	200 x 30 = 6000

Input	Value	Reason / Assumption
Registered users expected		
Active user percentage		
Peak factor		
Actions per user per minute		
Calculated RPS		
Daily active users		
Daily requests		

Activity 4: CPU Calibration

CPU sizing depends on request rate and processing complexity. Use the score below as a guided estimate.

CPU Driver	Score
Base web/API application	1
Authentication and role management	+1
Moderate API traffic above 5 RPS	+1
Database-heavy search or reporting	+1 to +2
Background jobs or notifications	+1
Analytics or dashboards	+2
AI/Python inference inside server	+2 to +4
Total CPU Score	Recommended vCPU
1-2	1-2 vCPU
3-4	2 vCPU

5-6	4 vCPU
7-9	4-8 vCPU or separate analytics/AI service
10+	Scale out with multiple VMs or managed compute

CPU Driver Selected	Score	Reason
Base web/API application		
Authentication and roles		
API traffic		
Database/search/reporting		
Background jobs		
Analytics/dashboard		
AI/Python module		
Total CPU Score		
Recommended vCPU		

Activity 5: RAM Calibration

RAM must include operating system, runtime, application process, database/cache and a safety buffer.

RAM Formula Estimated RAM = OS/runtime + app/API + database/cache + analytics/AI + 25% buffer. If database is managed separately, do not include full database memory in the app VM.		
Component	Guidance	Student Estimate
OS + runtime	1-2 GB	
Web/API service	1-2 GB	
Small database	2-4 GB	
Medium database/cache	4-8 GB	
Analytics/reporting	4-8 GB extra	
AI/Python module	4-16 GB extra	
25% buffer	Subtotal x 0.25	
Final RAM	Round up to standard size	

Activity 6: Storage Calibration

Storage must cover application files, database records, uploads, logs and backup/growth buffer.

Storage Item	Formula / Guidance	Student Estimate
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OS and application	20-40 GB	
Database	Avg record KB x records/day x days x 1.3 / 1024 MB	
File uploads	Avg file MB x files/day x days x 1.3	
Logs	Requests/day x log KB x days / 1024 MB	
Backup/growth buffer	Subtotal x 30%	
Final storage	Round up to standard disk size	

Technical Tip

Do not store large uploads permanently inside the VM disk when object storage is available. VM disk is for OS and application. Object storage is better for images, PDFs, media and documents.

Activity 7: Select the VM Plan

Plan	When It Fits	Student Choice
Single VM prototype	Small demo, simple app, low traffic	
Two-tier design	App VM + managed/separate database	
Three-tier design	Frontend, backend/API and database separated	
Containerized backend	Microservices, portable API, CI/CD	
Serverless function	Event-based task, notification, small automation	
Managed database/storage	Reliability, backup and scaling required	
Final VM Plan: selected plan, vCPU, RAM, storage and reason		

Activity 8: Hypervisor and Virtualization Decision

Approach	Meaning	Best Use
Type 1 hypervisor	Runs directly on physical hardware	Production data center and cloud provider infrastructure
Type 2 hypervisor	Runs above an existing OS	Student lab, local testing, VirtualBox/VMware Workstation
Cloud-managed virtualization	Cloud provider hides hypervisor details	Most public cloud VM deployments

Container runtime	OS-level process isolation	Lightweight APIs, microservices, fast deployment
Context	Selected Approach	Justification
Local prototype/testing		
Cloud pilot deployment		
Production growth scenario		
Module suitable for container		

Activity 9: Snapshot and Clone Strategy

Snapshot protects a point in time. Clone creates a duplicate environment. Both are useful, but they are not the same.

Event	Snapshot or Clone?	Frequency / Timing	Retention / Reason
Before major deployment			
Before database schema change			
Before OS/package update			
Before review/demo			
Create testing environment			
Create team demo copy			

Technical Tip

A snapshot is useful for rollback. A clone is useful for parallel testing. A clone should not be treated as the only backup.

Activity 10: VM vs Container Decision

Criteria	Virtual Machine	Container
Isolation	Strong OS-level isolation	Process-level isolation
Startup	Slower	Faster
Resource use	Higher	Lower
Best for	Full OS, legacy app, database VM	Microservices, APIs, portable services
Portability	Moderate	High
Student use	Clear infrastructure learning	Modern deployment learning
Capstone Module	VM / Container / Managed Service	Reason
Frontend		
Backend/API		
Database		
File storage		

AI/Python module		
Analytics/dashboard		
Notification/background jobs		

Activity 11: Final Recommendation

Write the final infrastructure recommendation in 8-12 technical lines. Include VM size, hypervisor/virtualization approach, storage plan, snapshot/clone plan and VM-container decision.

Carry-Forward to CO2 AT2 and CO2 AT3

Output from CO2 AT1	How It Will Be Used Later
VM sizing values	Used as configuration target in CO2 AT2 practical evidence
Hypervisor/virtualization choice	Used to justify deployment environment
Snapshot/clone plan	Used as backup and testing evidence
VM/container comparison	Used in CO2 AT3 infrastructure design case
Final recommendation	Used as the capstone infrastructure baseline

GitHub Submission Checklist

- Completed workbook file.
- README with capstone title, sizing summary and final recommendation.
- Optional architecture diagram or table exported as image/PDF.
- Repository link submitted in Google Classroom.